Topic Modelling with BERT

April 3th 2026

Session 1

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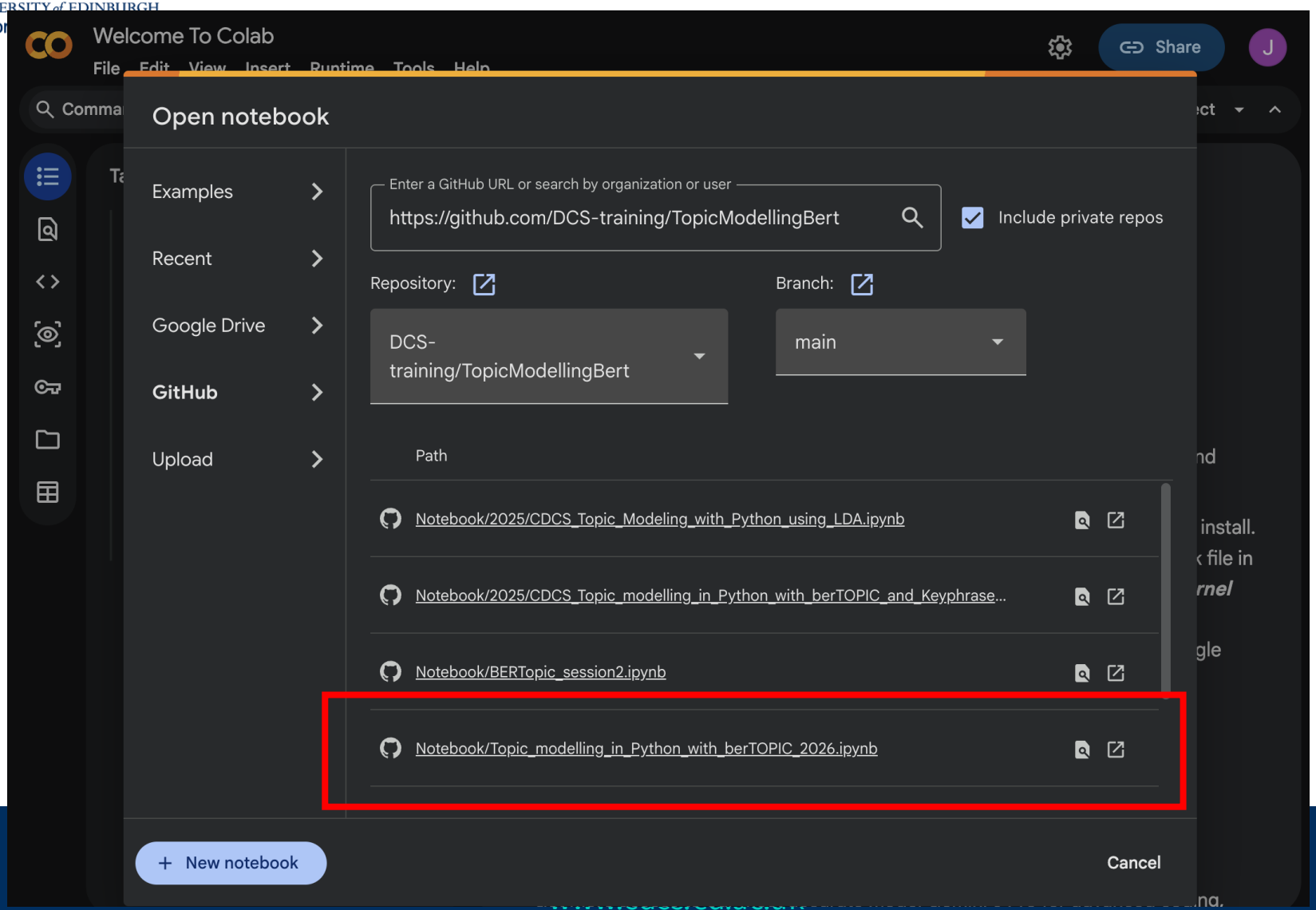
Course plan

W1: Lead by **Joy Lan**

What is topic modelling
What is BERT
How does BERT work

W2: Lead by **Aybuke Atalay**

Interpretation
Visualisation
Fine-tuning


Welcome To Colab
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Enter a GitHub URL or search by organization or user
https://github.com/DCS-training/TopicModellingBert
Include private repos
Repository:
DCS-training/TopicModellingBert
Branch:
main
Path
Notebook/2025/CDCS_Topic_Modeling_with_Python_using_LDA.ipynb
Notebook/2025/CDCS_Topic_modelling_in_Python_with_berTOPIC_and_Keyphrase...
Notebook/BERTopic_session2.ipynb
Notebook/Topic_modelling_in_Python_with_berTOPIC_2026.ipynb
+ New notebook
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Noteable

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Notebook

Python 3 (ipykernel)

Clone a repo

Enter the URI of the remote Git repository ning/TopicModellingBert

☑ Include submodules
☐ Download the repository

Cancel Clone

Billions of texts. What are they saying?

Every day we generate more text than any person can read.

Journals, policy documents, reports, news, social media…

We want to (quickly) discover what people are talking about?

Where do you start?
How can we make sense of the text?

Know your data, know your goal

Q1. What do you know about your data?

- Metadata: source, timeframe, language, how it was collected?..
- Content: general topic, coherence, length, etc.

Q2. What is your analytical goal?

- Exploring patterns in a collection of documents
- Classify documents into categories
- Analysing meaning, framing and discourse in texts

Supervised & Unsupervised learning in NLP

Supervised learning such as text classification, can be used to:

- Train a model that can later detect or predict a “defined label”.
- Validate and test coding scheme.

You’ll need:

- A defined label set
- **Labelled** documents
- Model configurations

You’ll get:

- A trained model that can assign predicted label(s) to new data
- Evaluation metrics that help you assess the model performance

Supervised & Unsupervised learning in NLP

Unsupervised learning , such as topic modelling can be used to:

- Train a model that can later detect or predict a “defined label”.
- Validate and test coding scheme.

You’ll need:

- An optional label set
- **(Unlabelled)** documents
- Model configurations

You’ll get:

- Topics/clusters described by keywords/representative texts
- Topic–document relationships

When do you use topic modelling in NLP?

Goal	Input	Output	Choice
• Detection and prediction • Validate and test coding scheme	• A defined label set • **Labelled** documents • Model configurations	• Predicted label(s) for new documents • Evaluation metrics on test data	supervised learning
• Discover themes/patterns • Generate candidate codes for qualitative analysis • Support sense-making and sampling for close reading	• A defined label set (opt.) • **Unlabelled** documents • Model configurations	• Topics/clusters described by keywords/representative texts • Topic–document relationships	unsupervised learning Topic modelling

Machine learning methods cannot replace human interpretation.

What we can and cannot do with topic modelling?

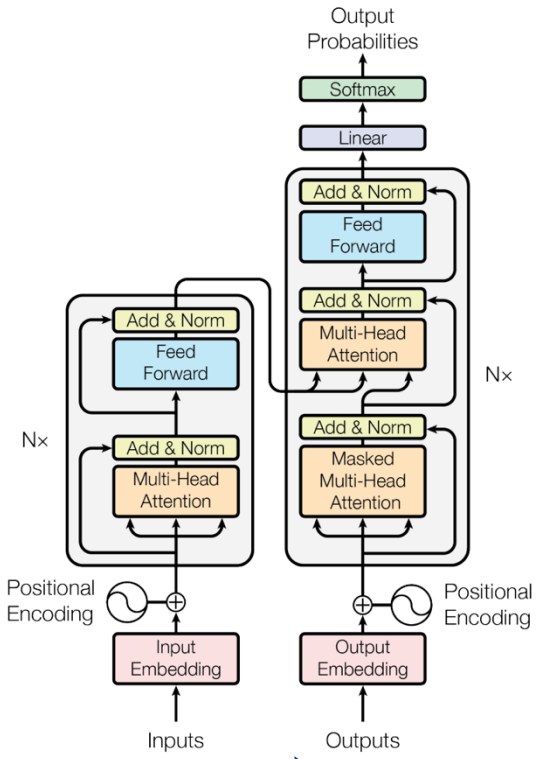
1. What themes appear in collection of documents?
2. How do discussions about a specific issue change over time?
3. How do different organisations emphasise different issues?
4. What is the author's main argument in this document?
5. Is this text supportive or critical of a policy/idea?
6. What causes a particular issue or outcome?

How does topic modelling work?

BERT - Transformer models

BERT

Encoder



GPT

Decoder

Transformer Model

Embedding + Positional Encoding + self attention

- **Embedding** turns each token in a piece of text corresponds to a vector in semantic space
- **Positional encoding** generate another vector that represents the position (order) of the word in the text.
- **Self-attention** is relational weighting, and the results reweighted the embedding vector.

- ex. The animal didn't cross the street because it was too tired.

BERTopic – Configuration

Steps	What does it do?	Syntax
01 Text representation	Turn each document into numerical form that captures sematic meanings.	embedding_model = "all-mpnet-base-v2”
02 Reduce dimensionality	Makes clustering more effective Shapes how topics will form	umap_model (applied by default)
03 Cluster reduced embedding	Groups documents into clusters based on similarity	hdbscan_model (applied by default)
04 Topic representation	Extracts words/phrases from documents in each cluster Counts their frequency	vectorizer_model
05 Create topic	Identify words that are: frequent in this topic rare in other topics	ctfidf_model (class-based TF-IDF, applied by default)
06 Refine topic representation (optional)	Improves topic labels using embedding-based methods	representation_model KeyBERTInspired etc.

BERTopic – 01 Text Representation

- BoW: What words are there?
- TF-IDF: Which words matter most?
- BERT: What does the word mean here?

Text Representation - BoW

Corpus containing 2 texts:

Text 1:
I went to the **bank** to check my **interest** and a **charge** on my account, but the **current** balance looked wrong.

Text 2 :
The **current** in the **stream** created a steady **flow** of water, while the **battery** powered a **light** in the open **field** near the **bank**.

Word	T1	T2
bank	1	1
interest	1	0
charge	1	0
current	1	1
battery	0	1
light	0	1
stream	0	1
field	0	1
flow	0	1

Text Representation - TF-IDF

Corpus containing 2 texts:

Text 1:
I went to the **bank** to check my **interest** and a **charge** on my account, but the **current** balance looked wrong.

Text 2 :
The **current** in the **stream** created a steady **flow** of water, while the **battery** powered a **light** in the open **field** near the **bank**.

Word	T1	T2
bank	0.2	0.2
interest	0.9	0.0
charge	0.9	0.0
current	0.2	0.2
battery	0.0	0.9
light	0.0	0.9
stream	0.0	0.9
field	0.0	0.9
flow	0.0	0.9

Text Representation – Embedding Model

Corpus containing 2 texts:

Text 1:
I went to the **bank** to check my **interest** and a **charge** on my account, but the **current** balance looked wrong.

Text 2 :
The **current** in the **stream** created a steady **flow** of water, while the **battery** powered a **light** in the open **field** near the **bank**.

Word	T1	T2
bank	[0.8, 0.7, 0.1]	[0.2, 0.3, 0.6]
current	[0.7, 0.6, 0.2]	[0.3, 0.4, 0.2]
interest	[0.9, 0.8, 0.1]	—
stream	—	[0.2, 0.3, 0.9]
battery	—	[0.3, 0.5, 0.8]

BERTopic – 02 Reduce Dimensionality

Define how documents cluster: UMAP

- Embeddings are high-dimensional (each token is a vector)
- UMAP reduces them and it allows easier clustering

- You can determine if you want to focus on local structure(more topics) or global structure (fewer topics)

BERTopic – 03 Cluster reduced embedding

Groups documents into clusters: HDBSCAN

- It is an unsupervised machine learning algorithm used for clustering, based on similarity.
- This is where topics are generated

- You can determine the number of clusters you want and adjust the algorithm

BERTopic – 04 Topic Representation

Extracts words/phrases from documents in each cluster: topic vectorizer

- This determines how your topic is represented/described.

- You can remove overly common and rare words
- Use multi-word expressions instead of one word (N-gram)

BERTopic – 05 Create Topics

Create the topics: c-TF-IDF

Now that the documents are grouped together into clusters.
This step treats each cluster as one document

Identify words that are:
frequent in this topic and rare in other topics

Topic modelling: workflow

a statistical method to discover topics in a set of documents

1. Cleaning
2. Tokenisation
3. Build model
4. Train model
5. Evaluation + visualisation

Let’s Code

Session 1

Codebook

Text cleaning

4035				en	x1163435477392867328	On #whd2019, join the @UN in celebrating #WomenHumanitaria
4036				en		
4037				en		
4038		x70985635	UN_HRC	en		
4039						en
4040				x1016	UN_SPExperts	en
4041				x1415	UN	en

The original data set has a mismatch of columns at row 4037

Codebook

Text cleaning

```
for t in en_data["text"]:
    cleaned = strip_emoji(t)                # 1. remove emojis
    cleaned = strip_all_entities(cleaned)   # 2. remove URLs, mentions, punctuation
    cleaned = clean_hashtags(cleaned)       # 3. clean hashtag patterns
    cleaned = filter_chars(cleaned)         # 4. remove unwanted symbols ($, &)
    cleaned = remove_mult_spaces(cleaned)   # 5. collapse multiple spaces
    clean_texts.append(cleaned)
```

Codebook

Embedding model

```
# 01 - Create embeddings
embedding_model = "all-mpnet-base-v2"
```

"all-mpnet-base-v2" is a pre-trained Sentence Transformer (based on BERT-like architecture)

It converts each document into a vector (numbers) or simply put, it turns each document into one dense vector

See: https://huggingface.co/sentence-transformers

Codebook

UMAP (default)

```
umap_model = UMAP(
    n_neighbors=15,
    n_components=5,
    min_dist=0.0,
    metric='cosine',
    random_state=42
)
```

n_neighbors

Low → captures **local structure**

High → captures **global structure**

n_components

Dimensions after reduction

min_dist

Low → tighter clusters

High → more spread-out clusters

random_state

Allows you to reproduce the same results

Codebook

HDBSCAN (default)

```
# 03 - Cluster reduced embeddings
hdbscan_model = HDBSCAN(
    min_cluster_size=15,
    metric='euclidean',
    cluster_selection_method='eom',
    prediction_data=True)
```

min_cluster_size

Low → fewer topics

High → more topics

metric

How distance between documents is measured

cluster_selection_method

eom = Excess of Mass

Chooses **stable, dense clusters**

Alternatively use 'leaf' for smaller clusters

Codebook

KeyphraseVectorizers

```
topic_model = BERTopic(embedding_model="all-mpnet-base-v2",
                       language="english",
                       calculate_probabilities=True,
                       verbose=True,
                       vectorizer_model=KeyphraseCountVectorizer(max_df = int(0.5*len(docs)),
                                                                 min_df = 5, decay = 0.1, delete_min_df=5.0)
                       )
```

KeyphraseCountVectorize extract phrases that describe the topics
c-TF-IDF is automatically applied during this step

max_df = 0.5 * len(docs) → Ignore phrases appearing in **too many documents**
min_df = 5 → Only keep phrases appearing in at least 5 documents
decay = 0.1 → Reduces influence of very frequent terms over time
delete_min_df = 5.0 → Removes very rare/unstable phrases

Codebook

Calculate probabilities

```
topic_model = BERTopic(embedding_model="all-mpnet-base-v2",
                       language="english",
                       calculate_probabilities=True,
                       verbose=True,
                       vectorizer_model=KeyphraseCountVectorizer(max_df = int(0.5*len(docs)),
                                                                 min_df = 5, decay = 0.1, delete_min_df=5.0)
                       )
```

Documents can partially belong to multiple topics.

Let’s Code

Session 1

Codebook

Get topic information

```
topic_model.get_topic_info()
```

	Topic	Count	Name	Representation	Representative_Docs
0	-1	2106	-1_human rights_human_rights_amp	[human rights, human, rights, amp, standup4hum...	[we hold up a human rights mirror to governmen...
1	0	169	0_myanmar_rohingya_bangladesh_military	[myanmar, rohingya, bangladesh, military, retu...	[myanmar alongside episodic violence a campaig...
2	1	122	1_lgbti_lgbti people_love_lgbt	[lgbti, lgbti people, love, lgbt, identity, se...	[stand up for equal rights amp fair treatment ...
3	2	112	2_universal declaration_declaration_udhr70_art...	[universal declaration, declaration, udhr70, a...	[watch live celebrate the 70th anniversary of ...

Each topic is represented by a group of words
-1 refers to all outliers and should typically be ignored.

Codebook

Get topic words

```
topic_model.get_topic(topic=5)

[('climate', np.float64(0.07330344140298843)),
 ('climateaction', np.float64(0.05783934058155271)),
 ('climatechange', np.float64(0.05170335324486828)),
 ('climate change', np.float64(0.045480836287091354)),
 ('change', np.float64(0.035214848611070555)),
 ('climate action', np.float64(0.02749428536810015)),
 ('climate crisis', np.float64(0.01874932097541701)),
 ('impacts', np.float64(0.01668726185312584)),
 ('action', np.float64(0.015044765487555971)),
 ('need', np.float64(0.011944187843941691))]
```

Each topic is composed of a group of words.
Words with higher scores mean that the word are more representative of the topic, compared to the other words.

Codebook

Get representative documents

```
topic_model.get_representative_docs(5)
```

```
['if you are feeling the impacts of climate change in your daily life yo
climatechange tell us below adaptourworld climateaction',
 'why is chief at cop24 because climate change is a terrible threat to h
 'we supports the right here right now global climate alliance in callin
treat climatechange as the humanrights crisis it is learn more about rhr
```

This function returns the documents sample that best embody the semantic meaning of the topic.

Codebook

Check the topic of each document

```
topic_model.topics_[:10]
```

```
[9, -1, 22, 23, 23, 23, 9, -1, -1, -1]
```

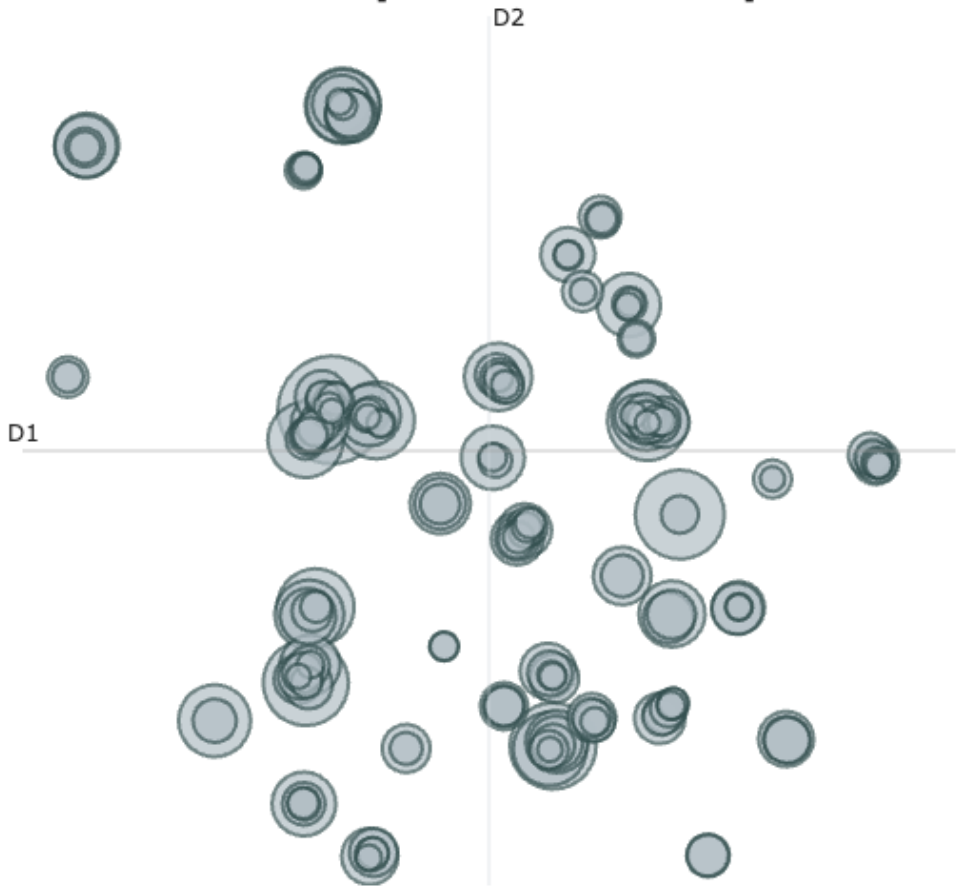
In this example,
Document 1 belongs to topic 9,
Document 2 does not have a main topic, etc.

Attributes

- There are a number of attributes that you can access after having trained your BERTopic model:

Attribute	Description
topics_	The topics that are generated for each document after training or updating the topic model.
probabilities_	The probabilities that are generated for each document if HDBSCAN is used.
topic_sizes_	The size of each topic
topic_mapper_	A class for tracking topics and their mappings anytime they are merged/reduced.
topic_representations_	The top *n* terms per topic and their respective c-TF-IDF values.
c_tf_idf_	The topic-term matrix as calculated through c-TF-IDF.
topic_labels_	The default labels for each topic.
custom_labels_	Custom labels for each topic as generated through `set_topic_labels`.
topic_embeddings_	The embeddings for each topic if `embedding_model` was used.
representative_docs_	The representative documents for each topic if HDBSCAN is used.

Intertopic Distance Map



Topic 0

Session 1 Wrap-up

Next Week:

- How can we interpret the results?
- How can we improve the model performance (that make more sense)

Additional Resources

examples of supervised learning in social studies

- Nelson, L. K., Burk, D., Knudsen, M., & McCall, L. (2021). The future of coding: A comparison of hand-coding and three types of computer-assisted text analysis methods. *Sociological Methods & Research*, *50*(1), 202-237.

- Rona-Tas, A., Cornuéjols, A., Blanchemanche, S., Duroy, A., & Martin, C. (2019). Enlisting supervised machine learning in mapping scientific uncertainty expressed in food risk analysis. *Sociological Methods & Research*, *48*(3), 608-641.

Additional Resources

examples of topic modelling

- Grubbs, A. E., Sinha, N., Garg, R., & Barber, E. L. (2023). Use of topic modeling to assess research trends in the journal Gynecologic Oncology. *Gynecologic oncology*, *172*, 41-46.

- Grajzl P, Murrell P. Using Topic-Modeling in Legal History, with an Application to Pre-Industrial English Case Law on Finance. *Law and History Review*. 2022;40(2):189-228. doi:10.1017/S0738248022000153

Thank you